Zuoming Niu

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EDUCATION

Xi'an Jiaotong University | *Faculty of Electronic and Information Engineering* Sept. 2022- Jun. 2026

- **Degree:** BS in Computer Science and Technology (Special Class for Integrated Circuits) **GPA:** 87.13/100

National University of Singapore | *School of Computing Workshop* May 2025-Jul. 2025

- **Program Overview:** Completed an intensive Computer Security research workshop, collaborating with a 5-member team on a federated learning privacy project that earned the course's Third Prize.

PROJECT EXPERIENCE

FL Breaker: Security Weakness of Federated Learning | *Leader* May 2025-Jul. 2025

Supervised by Visiting Associate Professor Hugh Anderson from National University of Singapore

- Led DLG attack reproduction, built on GitHub open-source implementations (MIT-HAN Lab's DLG), ran it on standard datasets, measured reconstruction quality, and built live loss-curve plots that show private images emerging from random seeds
- Mapped DLG pain points, heavy compute, joint (x', y') opt, local minima, and explained how IDLG bypasses them with label-only gradient trick and lighter optimizer
- Produced key visuals (Figure-style recovery snapshot and evolution sequence) that illustrate noise-to-image morphing during gradient matching, and curated a privacy–utility–cost comparison of HE/SMC/DP defenses
- Co-authored a comprehensive survey analyzing confidentiality, integrity, and availability attacks/defenses in federated learning; the work served as the course's final research deliverable and received the Third Prize

VLSI Design Lab of Xi'an Jiaotong University | *Lab Member* Oct. 2024-Dec. 2024

Supervised by Professor Hong Zhang from Xi'an Jiaotong University

- Co-developed a full-custom 8-bit addressable latch SN74HC259 in 0.18 μm CMOS with a 3-engineer team, owning the complete circuit-to-layout flow
- Drafted gate-level schematics for D-latch, inverter, NAND/NOR, buffer, and I/O cells in Cadence Virtuoso; verified V_IL/V_IH, V_OH/V_OL, T_pd, and pulse width through pre-layout Spectre simulations
- Implemented partial custom layout (2-metal layers) under strict area constraints; optimized cell placement and routing to comply with SMIC PDK design rules, achieving a DRC/LVS-clean layout
- Performed parasitic extraction and post-layout SPICE simulations (Cadence Spectre) to verify timing (T_su, T_h, T_pd, T_w) and average power, confirming all design targets were met and specifications satisfied
- Managed project timeline, completed complex tasks within deadlines, and communicated to optimize routing solutions for layout issues, ensuring circuit performance and enhancing coordination and systematic work skills

EXTRACURRICULAR ACTIVITIES

Xi'an Jiaotong University Student TV Station | *Station Director* Jul. 2024-Jul. 2025

- Directed an 80+ member student media team, leading full-stack production of large-scale live broadcasts for XJTU's major campus festivals and public events (orientation and graduation galas)
- Operated XJTU's official TikTok (1.3 M followers) and auxiliary accounts; produced 40+ videos and graphic posts in one year, accumulating 100k+ likes and sustaining audience growth
- Awarded 2024 "Outstanding Student Journalist" among Chinese universities; upgraded campus TV station from three-star to four-star society based on performance metrics

HONORS & AWARDS

- First-class Scholarship of Xi'an Jiaotong University 10/2025
- Excellent Student of Xi'an Jiaotong University 10/2025
- 2025 Mathematical Contest in Modeling Honorable Mention 01/2025
- First Prize in the 4th "My Xi'an Jiaotong University" English Video Competition 11/2024
- Third-Class Scholarship of Xi'an Jiaotong University 10/2023;10/2024
- First Prize in the Scholarly Reading Contest 04/2024
- Outstanding Student Cadre of Xi'an Jiaotong University 10/2023

SKILLS

- **Technical Skills:** Proficient in C++, Python, MS Office, Hspice, Cadence, MATLAB, etc.
- **Language Skills:** Chinese (Native); English (Fluent, TOEFL 103)